



ABCD Campus — Complete TWA & Google Play Store Publishing Guide

Authoritative, Step-by-Step Blueprint Tailored Specifically to the ABCD Smart Campus Architecture

Everything you need to know about building, publishing, auto-updating, background notifications, and install banners.



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1. Deep Scan Findings

A thorough scan of your codebase confirms that **95% of the technical foundation is already complete:**

Component	Status in ABCD Project	File Location
Web Manifest	✔ Complete with 96px, 192px, 512px maskable icons, standalone display mode, purple theme colors, and <code>in.abcdcampus.app</code> package ID	[site.webmanifest] (file:///b:/ABCD/abcd_web/static/data/favicon/site.webmanifest)
Service Worker	✔ Full push listener, custom audio playback, actions (Stop/Snooze/Open), notification grouping, and background alarm dispatch	[sw.js](file:///b:/ABCD/abcd_web/static/sw.js)
Digital Asset Links View	✔ Pre-configured endpoint serving <code>/.well-known/assetlinks.json</code> with package <code>in.abcdcampus.app</code> and configurable SHA-256 fingerprint	[users/views.py] (file:///b:/ABCD/abcd_web/users/views.py#L16895)
Asset Links Route	✔ Mapped in root URLs	[abcd_web/urls.py] (file:///b:/ABCD/abcd_web/abcd_web/urls.py#L48)
Smart App Detection	✔ Sophisticated logic with <code>getInstalledRelatedApps()</code> , Android Intent (<code>intent://...#Intent;package=in.abcdcampus.app;end</code>), and protocol fallback	[app-install-prompt.js](file:///b:/ABCD/abcd_web/static/js/app-install-prompt.js)
Smart Banner & Modals	✔ Sleek top banner and VIP learning experience modal with benefits	[app-install-prompt.js](file:///b:/ABCD/abcd_web/static/js/app-install-prompt.js)
Push Backend (VAPID)	✔ <code>send_push</code> pipeline with <code>sound</code> , <code>is_alarm</code> , <code>is_reminder</code> , and custom payload fields	[users/notifications.py] (file:///b:/ABCD/abcd_web/users/notifications.py)
High-Res App Icons	✔ All present in favicon folder: <code>favicon-96x96.png</code> , <code>web-app-manifest-192x192.png</code> , <code>web-app-manifest-512x512.png</code>	[static/data/favicon/] (file:///b:/ABCD/abcd_web/static/data/favicon)

2. TWA Fundamentals

Q: “If I change or fix anything on the site, do I need to update the APK/AAB on Google Play Store?”

[!IMPORTANT] NO! You DO NOT need to build a new APK/AAB or republish to Google Play Store.

Why does this happen? (How TWA Works)

- A traditional native app (built in Java, Kotlin, Flutter, or React Native) compiles code into a bundle. Every HTML/CSS/JS or logic change requires a new build and 2–3 days for Google review.
- A **Trusted Web Activity (TWA)** is Google Chrome’s official technology that wraps your live website in a borderless, full-screen Android app shell.
- When a student opens the ABCD app on Android, the shell **instantly loads** `https://abcdcampus.in` live from your Render server.
- **Result:**
 - Whenever you push code to GitHub and Render redeploy, **every single user’s app updates instantly in real time.**
 - Bug fixes, new features in Guidy, library seat updates, audio changes, and design adjustments are reflected immediately with zero download or update required from Play Store!

When WOULD you ever need to republish a new AAB to Play Store?

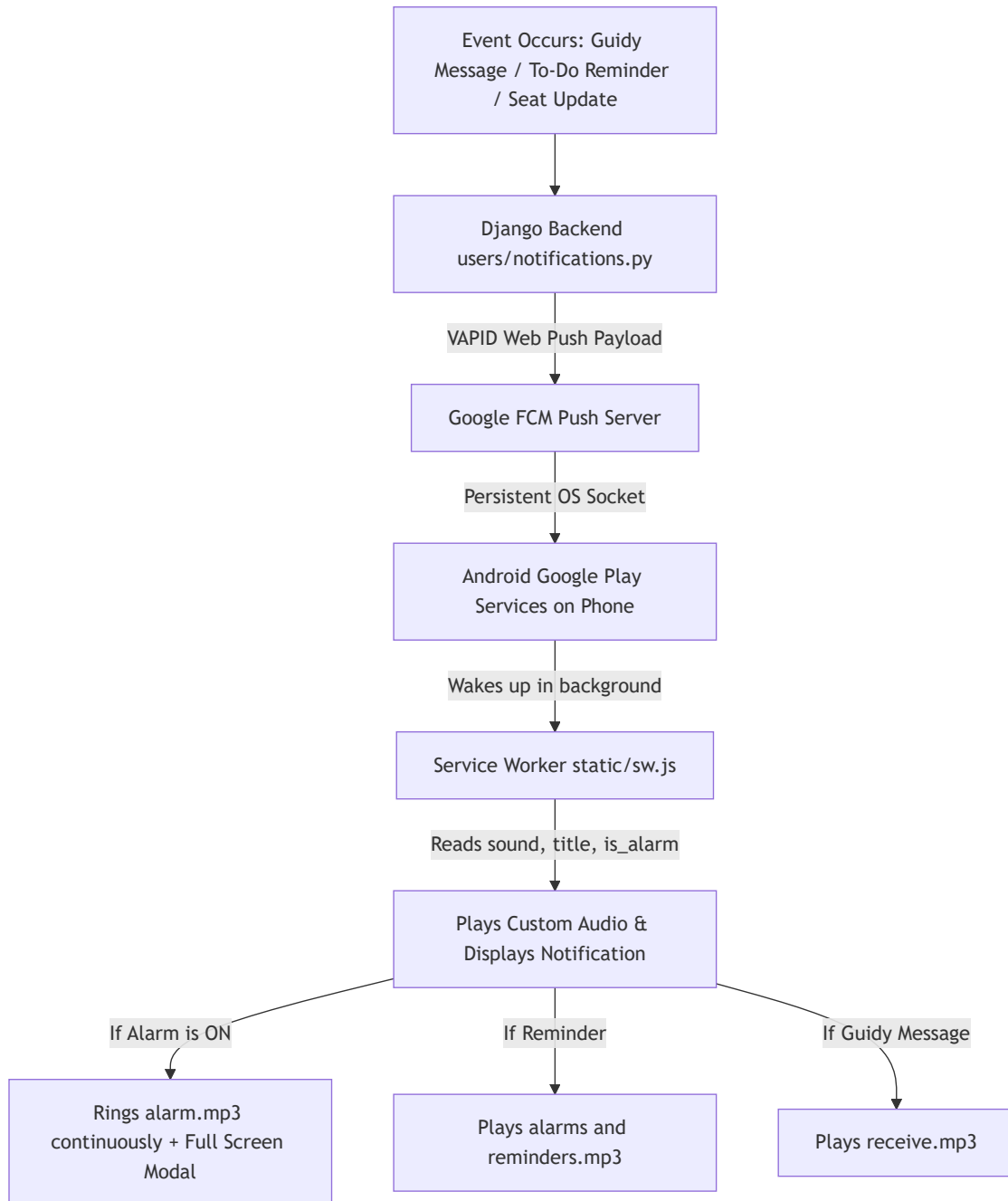
Only in these 3 rare cases:

1. You change the Android app icon that appears on the phone’s home screen.
2. You change the splash screen background color or app name in Android OS settings.
3. Once a year, Google Play requires bumping the target Android SDK version (e.g. from Android 14 to Android 15).

3. Background Notifications When Closed

Q: "How to ensure that even when tabs/browser are completely closed, Guidy messages, reminders, and alarms still ring on the device?"

The Delivery Flow:



Why It Works Even With Browser Closed:

1. The phone's operating system maintains a permanent, ultra-low-power network connection to Google's push servers via **Google Play Services**.
2. When your server sends a push notification, Google's servers wake up your Service Worker on the user's phone, even if Chrome, the browser, and the ABCD app are closed or killed from recent tasks.
3. The Service Worker receives the `push` event, reads the payload (`sound` , `is_alarm` , `category`), and calls `self.registration.showNotification()` .

Crucial Steps on the User's Phone:

Modern Android versions (Android 13+) and aggressive phone skins (MIUI/Xiaomi, Realme, Vivo, Oppo, Samsung) have aggressive battery optimizations. To ensure 100% instant ring:

1. **Notification Permission:** Android 13+ asks: "Allow ABCD Campus to send notifications?" The user must tap **Allow**.

2. **Battery Optimization (For Alarms):** In phone *Settings* → *Apps* → *ABCD Campus* → *Battery*, recommend setting to “Unrestricted”.
3. **Auto-Start (MIUI/Vivo/Oppo):** Enable “Autostart” so background alarms fire on schedule even after device reboot.

4. Step-by-Step Generation

You need two files:

- **.apk** : For installing directly on your Android phone to test immediately before publishing.
- **.aab** (Android App Bundle): The exact format required by Google Play Console for store upload.

Method 1: PWABuilder (Recommended — Zero Java/SDK Setup)

PWABuilder is Microsoft's official open-source tool that uses Google's Bubblewrap on cloud build servers. You don't need to install Java, Android Studio, or Android SDK on your computer.

1. Open <https://www.pwabuilder.com> (<https://www.pwabuilder.com>) in your browser.
2. Enter your live domain: `https://abcdcampus.in` and click **Start**.
3. PWABuilder will audit your PWA (Manifest, Service Worker, Security) — you will get a high score because your manifest and service worker are already compliant.
4. Click **Package for Stores** → Choose **Android** → Click **Generate**.
5. In the Android Package Options dialog, fill in:
 - **Package ID:** `in.abcdcampus.app`
 - **App Name:** `ABCD Coaching & Library`
 - **Short Name:** `ABCD Campus`
 - **Version Name:** `1.0.0`
 - **Version Code:** `1`
 - **Display Mode:** `Standalone`
 - **Status Bar Color:** `#e09fd5`
 - **Nav Bar Color:** `#854c7a`
 - **Splash Screen Color:** `#854c7a`
 - **Notifications:** Check **Enable Web Push Notifications** and **Notification Delegation**.
6. **Signing Key:**
 - Choose **Create New Keystore**.
 - Fill in your details (Alias: `abcd` , Password: create a strong password).
 - **[!CAUTION]** Download and save this `.keystore` file and its password in Google Drive or a safe backup! You need this key whenever you want to update the app in the future.
7. Click **Generate Bundle**.
8. PWABuilder will compile and provide a `.zip` download containing:
 - `app-release-bundle.aab` (upload this to Play Store)
 - `app-release.apk` (install this on your phone to test)
 - `assetlinks.json` (contains your SHA-256 fingerprint!)

Method 2: Local Bubblewrap CLI

If you prefer running the build locally on your Windows command prompt:

1. **Install Java JDK 17:**
 - Download from [Adoptium Temurin 17](https://adoptium.net/temurin/releases/?version=17) (<https://adoptium.net/temurin/releases/?version=17>) or install via PowerShell:

```
winget install EclipseAdoptium.Temurin.17.JDK
```

2. **Install Bubblewrap globally:**

```
npm install -g @bubblewrap/cli
```

3. Initialize the TWA:

```
mkdir b:\ABCD\abcd-twa
cd b:\ABCD\abcd-twa
bubblewrap init --manifest=https://abcdcampus.in/static/data/favicon/site.webmanifest
```

(Bubblewrap will auto-download the Android command-line SDK tools and prompt you to create a keystore).

4. Build APK and AAB:

```
bubblewrap build
```

5. Google Play Console Upload Flow

Since your friend has granted user access to their Google Play Developer account on your email:

1. Log in to play.google.com/console (<https://play.google.com/console>) with your Gmail.
2. Click **Create App**:
 - **App name**: ABCD Coaching & Library
 - **Default language**: English (India) - en-IN
 - **App or game**: App
 - **Free or paid**: Free
 - Accept the Developer Program Policies.
3. **Dashboard Setup Tasks (Required by Google)**:
 - **Privacy Policy**: Provide URL <https://abcdcampus.in/privacy-policy> (or your platform terms).
 - **App Access**: Choose “All functionality is available without special access” or provide demo credentials (`demo_student` / `password`).
 - **Ads**: Select “No, my app does not contain ads”.
 - **Content Rating**: Fill in the questionnaire (Education / Utility → No violent or adult content).
 - **Target Audience**: Age 13 and above.
 - **Government Apps**: No.
 - **Financial Features**: None (or student fee receipt viewer).
4. **Store Listing**:
 - **Short description** (up to 80 chars): *Smart campus app for library seats, courses, fee tracking & Guidy chat.*
 - **Full description**: Highlight library availability, fee calendar, student-teacher chat, and study reminders.
 - **App Icon**: Upload `web-app-manifest-512x512.png` (512x512 PNG).
 - **Feature Graphic**: 1024x500 PNG banner image.
 - **Phone Screenshots**: Take 3–5 screenshots from your phone showing the dashboard, library seat matrix, Guidy chat, and student profile.
5. **Releases → Production / Closed Testing**:
 - Create a new release.
 - Upload the `.aab` file (`app-release-bundle.aab`).
 - Release name: `1.0.0` .
 - Click **Save and Review Release**.

6. Digital Asset Links & App Signing

[!WARNING] **The Most Common Mistake**: When you upload your `.aab` to Google Play, Google uses **Google Play App Signing**. Google re-signs your app with *their* private key. If Google’s SHA-256 fingerprint is not in `assetlinks.json`, the browser URL bar will show up at the top of your app!

How to Fix This in 2 Minutes:

1. In Google Play Console, navigate to **Setup → App Integrity**.
2. Under **App Signing**, find the section “**App signing key certificate**”.
3. Copy the **SHA-256 certificate fingerprint** (it looks like `12:34:56:78:AB:CD:...`).

4. In your Render Dashboard:

- Go to **Environment Variables**.
- Add/Update:

```
TWA_SHA256_FINGERPRINT=YOUR_COPIED_SHA256_HERE
```

- If you want both local test APKs and Play Store to work simultaneously, you can pass comma-separated fingerprints or update `users/views.py` to support multiple fingerprints.

5. Verify your endpoint in your browser:

```
https://abcdcampus.in/.well-known/assetlinks.json
```

- 6. Google will automatically verify the domain ownership within a few hours. Once verified, the URL bar disappears completely, giving users a 100% native full-screen experience.

7. Smart Banner & Install Strategy

Q: “What should happen when the user visits from a mobile browser? Should we prompt browser install or redirect to Play Store?”

What Big Tech Does (Twitter, LinkedIn, Flipkart, Reddit):

On Android mobile browsers:

1. Redirect to Google Play Store is the Golden Standard:

- In India, 95%+ of students prefer installing from the **Google Play Store** rather than a browser popup.
- Play Store provides the recognizable “Verified by Play Protect” seal, auto-updates, and instant credibility.

2. Behavior Matrix:

- **On Android Mobile:** When user taps **INSTALL** in your top smart banner or modal, redirect them directly to:

```
market://details?id=in.abcdcampus.app
```

(Fallback: `https://play.google.com/store/apps/details?id=in.abcdcampus.app`). This opens the Google Play Store app directly on their phone with a big green **Install** button!

- **If App is Already Installed on Phone:** Your code in `[app-install-prompt.js]`(`file:///b:/ABCD/abcd_web/static/js/app-install-prompt.js`) automatically detects it via `getInstalledRelatedApps()` and changes the banner button to **OPEN**. Tapping it launches the app immediately via the native Android Intent!
- **On iOS / iPhone:** Google Play Store doesn’t exist on iOS. The modal automatically displays the iOS Safari instructions: “*Tap Share → Add to Home Screen*”.

8. Installing on Desktops & Laptops

Q: “Can our app be installed on laptops and desktops too?”

[!NOTE] YES! On Windows, Mac, and Linux, your app installs as a Desktop PWA.

How Desktop Users Install ABCD:

1. When visiting `https://abcdcampus.in` on Google Chrome or Microsoft Edge on a laptop:

- A small “**Install ABCD Campus**” icon (an icon with a computer and down arrow) appears in the browser’s URL address bar on the right.
- Or, clicking **Install App** in the navigation menu triggers the native desktop install prompt.

2. Once installed:

- An **ABCD Coaching & Library shortcut** is added to the Windows Desktop and Start Menu.
- The app opens in its own dedicated, clean window without browser tabs or address bars.
- Supports native Windows notifications!

3. Chromebooks (ChromeOS):

- Chromebooks natively support the Google Play Store! Students on Chromebooks can download the exact same Play Store app.

9. Pre-Launch Checklist

- ☐ Verify `https://abcdcampus.in` Verify `https://abcdcampus.in` is running live on HTTPS with valid SSL.
- ☐ Test push notification on your own phone using the web version first.
- ☐ Generate `.aab` and `.apk` Generate `.aab` and `.apk` via PWABuilder (or Bubblewrap).
- ☐ Install the `.apk` Install the `.apk` on your phone via USB or WhatsApp/Drive to test UI feel.
- ☐ Create App in Google Play Console using your friend's developer account.
- ☐ Copy the SHA-256 fingerprint from Google Play Console **App Integrity** into `TWA_SHA256_FINGERPRINT` Copy the SHA-256 fingerprint from Google Play Console **App Integrity** into `TWA_SHA256_FINGERPRINT`.
- ☐ Submit App for review (Review typically takes 1 to 3 days).
- ☐ Once published, test downloading directly from Google Play Store!